

[DC] The Essentials of XR Prototyping for Non-Tech Professionals: Taxonomy and Low-Fidelity Tools to Aid Prototyping Decisions

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ABSTRACT

This paper delves into the complexities of Extended Reality (XR), exploring the challenges in XR design, directing user attention, managing cybersickness, and the overall integrative nature of these issues in the context of prototyping. We offer an overview of the current state of XR and identify key areas where novice and non-technical professionals face challenges. The paper concludes with a proposed research agenda focused on three main themes and potential research directions. This work aims to enhance the design process and facilitate rapid prototyping, enabling non-technical professionals to create more effective and engaging XR experiences.

Index Terms: Human-centered computing—Interaction design—Interaction design process and methods—Interface design prototyping; Human-centered computing—Human computer interaction

1 OVERVIEW OF THE PROBLEM

Different forms of Extended Reality (XR) experiences, in general, and recently the metaverse, have been a hot topic due to its potential applications in many industries such as the entertainment industry. While this interest started in games, XR has proven to be an interesting storytelling medium. This growth has increased the interest in designing for XR by novice and non-tech-savvy designers. Moreover, due to its multifaceted applications, XR experiences design is attracting many interdisciplinary researchers, designers, and artists. In most cases, these applications could be forgiving, however, some issues related to this medium could hinder the growth of such applications due to the way they affect experiences.

Designing XR experiences poses significant challenges for novice designers, particularly those who are not tech-savvy. In the interdisciplinary area of XR, they often struggle with their unfamiliarity with XR technologies, limited knowledge of design principles specific to XR, and the complexity of integrating technical aspects with creative content. Due to their limited knowledge, they often face difficulties in understanding the specific needs and preferences of XR users, and the issues presented by these media, like guiding attention and cybersickness, leading to sub-optimal user experiences. Therefore, the interdisciplinary nature of XR design requires collaboration between technical developers and content creators to ensure a seamless integration of technical aspects with creative content. As a result, effective prototyping tools are a must as a means of communication [3].

Prototyping allows designers to test and refine their ideas, identify potential challenges, and gather feedback from users before moving forward with full-scale development of an XR experience. It is an iterative process in which ideation, physicalization and storyboarding play significant roles as prototyping methods as per Microsoft's model [4] as shown in Figure 1. Based on this model, we focused on physical prototyping and storyboarding as two important methods to develop in our prototyping toolkit.

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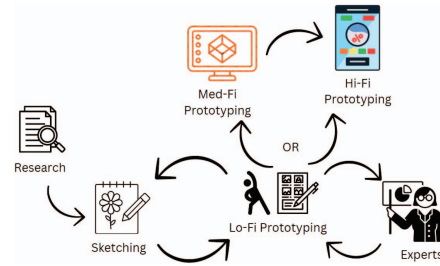


Figure 1: XR Prototyping based on Microsoft's model [4]

XR prototyping is an area we seriously lack in having enough tools to support the process [1]. A good prototyping tool removes technical constraints, that is why Low-Fidelity (Lo-Fi) prototyping tools like paper are great for XR [1]. This highlights the importance of design toolkits as promising knowledge mobility means between researchers and designers. Furthermore, most of XR prototyping tools have been designed for VR but not the whole continuum [1].

One of the major challenges that are unique to XR is guiding user attention [6]. This is caused by the interactivity and 360 surrounding nature of virtual environments. This issue is significant in storytelling as much as in utilitarian applications. Some of the current solutions use techniques such as sound cues, and these are not necessarily efficient when used in combination with other types of techniques such as visual cues. For example, they could introduce side effects such as cybersickness - motion sickness due to visual-vestibular conflicts during VR use [5]. This is significant because cybersickness has been proven to affect presence and cognitive performance negatively. Cybersickness Reduction Techniques (CRTs) in XR are important techniques to reduce cybersickness experienced in these interactive 360 displays. However, they tend to change the nature of the experience and could limit creative choices; and if cybersickness is not solved it could become an accessibility issue that would hinder the democratization of this technology. Additionally, there are some ethical issues to be considered, especially that cybersickness, presence, and user performance in XR are experienced differently based on biological gender-related factors [2], ethnicity, and other diversity-related factors [5].

It is evident that any XR experience constitutes a complex system. This complexity prompts the consideration of systems thinking as a potential solution. Norman's DesignX emphasizes the significance of incorporating interdisciplinary knowledge when navigating the intricacies of such complex, wicked systems, advocating a shift towards problem-based solutions. Nonetheless, the practicality of this approach remains uncertain in the absence of definitive disciplinary methodologies or, at the very least, a structured systematic approach to prevent it from devolving into mere speculations. It is increasingly recognized that comprehensive, holistic frameworks are imperative for managing these systems. Yet, within the XR domain, particularly in the context of emerging phenomena like the metaverse, there is still no consensus on the nature of these frameworks, their distinctiveness, or even if they differ fundamentally from existing ones.

2 RESEARCH DIRECTIONS

The primary objective of this research endeavour is to comprehensively investigate the potentialities of Lo-Fi prototyping, specifically storyboarding and physical modalities within the sphere of XR experiences, yet the focus will still be prototyping for attention guidance and cybersickness. We selected them due to their unique nature and significance. Directing attention facilitates the experience so we want to promote it. On the other hand, cybersickness hinders it through its symptoms, so we want to combat it, yet it is connected to the interaction technique used. We have conducted a thorough literature review to produce a comprehensive list of Attention Guidance Techniques (AGTs) and CRTs to be considered in prototyping tools.

We will conduct a series of experiments to address two interrelated problems a) providing an XR prototyping decision guidance taxonomy to aid novice and non-tech-savvy designers and artists in reducing errors in communicating the design attributes with the engineers, and b) assessing the role that Lo-Fi prototyping tools, namely storyboarding and physical prototyping, could play in addressing cybersickness and attention guidance design issues as two case studies. This includes considering any ethical implications of any proposed solutions through research findings and user feedback. We will be modifying and developing tools that allow creative professionals to create Lo-Fi prototypes that don't require technical skills and are rather focused on the creative process. Yet, these will facilitate prototyping for attention guidance, cybersickness, and 3D interactions. This will allow for lower technical skills entry-level by non-technical professionals.

This research methodology is based on conducting a series of experiments which will follow an iterative (design, develop, and evaluate) methodology. We are utilizing a mix of participatory research methods, in the form of co-design workshops, to generate tools that address these two issues, and user studies to verify the generated tools. These research studies are conducted remotely. For that purpose, we developed an informed remote user studies framework, which has been published as a journal paper. For better understanding, this research could be categorized under the three following themes.

2.1 Theme 1: Developing XR Prototyping Taxonomy for Decision Support

As part of this proposed prototyping framework and toolkit, this taxonomy facilitates an informed decision-making process to choose a suitable combination of prototyping methods at the beginning of the process. To produce this taxonomy, we synthesized insights from various comprehensive prototyping taxonomies in the contexts of XR and other disciplines to propose this taxonomy as a holistic decision-making aid. We further informed our taxonomy by surveying existing XR prototyping work in the literature. Intended primarily for novices and non-tech-savvy designers, our taxonomy could also serve as a valuable educational resource for introducing learners to XR prototyping nuances. Its first iteration is presented as a poster at IEEE VR 2024. We do not claim completion of the work but it is a significant step towards the desired outcome. Future work will focus on updating and automating this decision-making aid. Additionally, we will further evaluate its quality through a series of measures suggested by the literature, while its use will be evaluated through user studies.

2.2 Theme 2: Evaluation and Refinement of Storyboarding and Physical Prototyping Tools for XR

We will be conducting a series of co-design workshops to produce modified tools by stakeholders from interdisciplinary backgrounds and a variety of experience levels followed by user studies to evaluate the produced artifacts and tools. The tools will be storyboards, acting and bodystorming aids and recommendations, and 3D physical modelling recommendations. Additional inquiry will be assessing if

alternative tools, perhaps more inclusive than bodystorming, might be more effective given bodystorming's inherent biases towards able-bodied designers. Our exploration will be underpinned by a multi-faceted approach comprising questionnaires, participatory co-design studies, and semi-structured interviews.

In conclusion, these tools will be evaluated as a toolkit to ensure that they are achieving their desired outcome when used together. We foresee that they help in translating research findings and facilitate standardized rapid XR prototyping. We believe that such tools are highly needed due to the lack of prototyping methods that allow for rapid prototyping while considering AGTs or CRTs.

2.3 Theme 3: Knowledge Mobility of Novel Prototyping Techniques in the XR Research Community

In our efforts to mobilize this knowledge, we started conducting a series of conference workshops about XR prototyping (see www.xrprototyping.com). The first was hosted by IEEE VR 2023 and its second iteration is hosted by IEEE VR 2024. We aim for this workshop to become a permanent avenue for exchanging ideas, discussions, and research findings around this. These workshops facilitate turning our research findings into actions by the community.

3 CONCLUSION

This research will expand and improve the human resources available in this growing sector, we foresee that such research will enrich the field, encourage prototyping for attention guidance and cybersickness, lower the entry technical level to these prototyping tools and encourage a wider acceptance of the toolkit we propose.

In this paper, we provided an overview of the problem and the technical landscape. Then, we proposed a research plan and its potential directions. Through the three proposed themes, we introduce a comprehensive framework and toolkit aimed at enhancing the effectiveness of the XR prototyping process. This is achieved through informed decision-making, efficient low-fidelity prototyping, and the effective dissemination of research findings. To conclude, the three main areas I would like feedback on are: i) Are there additional theories that could help further ground my work? ii) How can we improve the design of the taxonomy? iii) Are there certain measures or experiments we should consider?

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